

Anibal Estuardo Sierra Morales

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Instituto de Astronomía - Universidad Nacional Autónoma de México - Ciudad de México

PROFESSIONAL EXPERIENCE

- **Associate Professor** Since November 2025
Instituto de Astronomía, Universidad Nacional Autónoma de México Ciudad de México
- **Postdoctoral researcher in Astrophysics** June 2024 - October 2025
Mullard Space Science Laboratory, University College London London, England
Working in the Planet Formation group of Dr. Paola Pinilla
- **Postdoctoral researcher in Astrophysics** April 2024 - May 2024
Pontificia Universidad Católica de Chile Santiago de Chile, Chile
Working in the group of Dr. Viviana Guzmán
- **Postdoctoral Fondecyt Fellow** April 2022 - March 2024
Agencia Nacional de Investigación y Desarrollo (ANID) Santiago de Chile, Chile
Working in the group of Dr. Laura Pérez
- **Postdoctoral researcher in Astrophysics** March 2020 - February 2022
Facultad de Ciencias Físicas y Matemáticas, Universidad de Chile Santiago de Chile, Chile
Working in the group of Dr. Laura Pérez

EDUCATION

- **Ph.D. in Astrophysics** January 2016 - December 2019
Instituto de Radioastronomía y Astrofísica - Universidad Nacional Autónoma de México Morelia, México
Thesis: Dust dynamics and emission in protoplanetary disks. Advisor: Dr. Susana Lizano
- **M.Sc. in Astrophysics** January 2014 - December 2015
Instituto de Radioastronomía y Astrofísica - Universidad Nacional Autónoma de México Morelia, México
Thesis: Dust dynamics in protoplanetary disks. Advisor: Dr. Susana Lizano
- **B.Sc. in Physics** January 2008 - December 2013
Facultad de Ciencias Físicas y Matemáticas - Universidad de San Carlos de Guatemala Guatemala, Guatemala
Thesis: Group representation in physics. Advisor: Ing. Rodolfo Samayoa

SELECTED HONORS AND AWARDS

- **Robert Boyd award** 2025
Space and Climate Physics, University College London United Kingdom
Logro científico destacado
- **Ph.D. thesis award** 2021
División de Astrofísica de la Sociedad Mexicana de Física México
Best Ph.D. thesis done in Mexico or by mexicans abroad (period 2020-21)
- **Alfonso Caso Medal** 2020
Universidad Nacional Autónoma de México México
Best student of the generation, Ph.D. in Astrophysics
- **Alfonso Caso Medal** 2015
Universidad Nacional Autónoma de México México
Best student of the generation, M.Sc. in Astrophysics
- **Cum Laude degree** 2013
Universidad de San Carlos de Guatemala Guatemala
B.Sc. in Physics
- **University student excellence award** 2013
Universidad de San Carlos de Guatemala Guatemala
Best student of the generation, Faculty of Engineering, Physics & Mathematics
- **Faculty student excellence award** 2009-2012
Universidad de San Carlos de Guatemala Guatemala
Best physics student of the year for 4 years

SUPERVISED THESIS

- **Advisor Bachelor Thesis** 2026-to date
Student: Litzy Valentina Guzmán Mayorquín. Universidad Autónoma de Sinaloa Ciudad de México, México
Thesis: "Evolución de la línea de nieve en discos protoplanetarios"
- **Advisor Bachelor Thesis** 2026-to date
Student: Renieri Ferrari. Universidad Autónoma de Honduras Tegucigalpa, Honduras
Thesis: "Propiedades físicas del disco alrededor de MWC 758"
- **co-Advisor M.Sc. Thesis in Astrophysics.** 2022-2023
Student: Luz Maritza Caballero Cano. Instituto de Radioastronomía y Astrofísica Morelia, México
Thesis: "Propiedades del polvo en el disco protoplanetario alrededor del sistema HD 142527"
- **Advisor M.Sc. Thesis in Physics.** 2022-to date
Student: Erick Alexander Rojas García. Universidad de San Carlos de Guatemala Guatemala, Guatemala
Thesis: "Impact of dust porosity on the radiative transfer properties in protoplanetary disks"

STUDENT SUPERVISION

- **Social Service Supervision (Universidad Nacional Autónoma de México)**
Students: Carlos Guillén (2026), Luis Gómez (2026)
- **Supervision of B.Sc. projects (Universidad Autónoma de Sinaloa)**
Student: Litzy Guzmán (2025)
- **co-Supervision of M.Sc. projects (Universidad Nacional Autónoma de México)**
Student: Luz Maritza Caballero (2022-2023)
- **Supervision of M.Sc. projects (Universidad de San Carlos de Guatemala)**
Student: Erick Rojas (2022-to date)
- **Supervision of B.Sc. projects (Universidad de Chile)**
Students: Benjamín Carrera (2023-2024), Benjamín Sotomayor (2022-2023)
- **Supervision of B.Sc. projects (Universidad de Chile)**
7 B.Sc. students between 2021-2024

TEACHING EXPERIENCE

- **Lecturer**
Graduate Program in Science (Astronomy) Ciudad de México, México
Radiative Processes in Astrophysics (Semester 2026-2)
- **Lecturer**
Preparatory courses, Graduate Program in Science (Astronomy) Ciudad de México, México
Classical Mechanics (Semester 2026-2, 2027-1)
- **Lecturer** 2020
Faculty of Physics and Mathematics, Universidad de Chile Santiago de Chile, Chile
General astronomy. Undergrad-level course for engineers and scientist students
- **Teaching assistant** 2019
Escuela Nacional de Estudios Superiores, Universidad Nacional Autónoma de México Morelia, México
Distributed computing. Undergrad-level course for science and technology students
- **Teaching assistant** 2018
Escuela Nacional de Estudios Superiores, Universidad Nacional Autónoma de México Morelia, México
Image analysis using Python. Undergrad-level course for science and technology students
- **Teaching assistant** 2013
Faculty of Engineering, Universidad Francisco Marroquín Guatemala, Guatemala
Calculus, linear algebra. Undergrad-level course for engineers students
- **Physics laboratory teacher** 2013
Physics Department, Universidad de San Carlos de Guatemala Guatemala, Guatemala
Classical Mechanics, Electromagnetism, and Optics. Undergrad-level laboratory for physics students

GRANTS AND SCHOLARSHIPS

- **FONDECYT fellow** 2022-2024
Agencia Nacional de Investigación y Desarrollo (ANID) Chile
- **CONACYT** 2016-2019
Consejo Nacional de Ciencia y Tecnología
Scholarship for Ph.D. studies México
- **CONACYT - OEA - AMEXCID** 2016
Organización de los Estados Americanos - Agencia Mexicana de Cooperación Internacional para el Desarrollo
Ph.D. travel and accommodation México
- **CONACYT - OEI** 2016
Organización de los Estados Iberoamericanos
Ph.D. travel and accommodation México
- **CONACYT** 2014-2015
Consejo Nacional de Ciencia y Tecnología
Scholarship for M.Sc. studies México
- **CONACYT - OEA - AMEXCID** 2014
Organización de los Estados Americanos - Agencia Mexicana de Cooperación Internacional para el Desarrollo
M.Sc. travel and accommodation México
- **USAC Grant** 2008-2013
Universidad de San Carlos de Guatemala
Grant for B.Sc. studies Guatemala

PROFESSIONAL SERVICE

- **per-Reviewer** Since 2021
Scientific articles in ApJ, ApJL, and A&A. ~ 2 papers per year
- **Secretary** 2020-2021
Central America and Caribbean association for Astronomy Sciences (Alpha-Cen)
- **Evaluation commission** 2012-2013
Guatemalan Basic Sciences Olympiad
- **ad-Honorem teacher assistant** 2011
Faculty of Engineering, Universidad de San Carlos de Guatemala

OBSERVATIONAL EXPERIENCE

- **Awarded observation time:** ALMA and VLA (as PI and Co-I), APEX, and JWST (as Co-I)
Full list of observational time at anibalsierram.github.io/Research.html

PRESENTATIONS AND SCIENTIFIC TALKS

- **Colloquium and Seminar speaker** at over 12 different universities and observatories.
- **Contributed speaker** to over 6 national and international conferences
Full list of Seminar and contributed speaker at anibalsierram.github.io/Talks.html

PUBLIC OUTREACH

- **Speaker at multiple events for general public.** Guatemala, Mexico, and Chile. ~ 30 events.
- **Scientific press releases for ALMA Large program papers.** alma-maps.info/press.html, and AGE-PRO website in progress.

REFERENCES

1. **Susana Lizano**
Professor, Instituto de Radioastronomía y Astrofísica
Universidad Autónoma de México
Email: s.lizano@irya.unam.mx
M.Sc. and Ph.D. advisor
2. **Laura Pérez**
Professor, Departamento de Astronomía
Universidad de Chile
Email: lperez@das.uchile.cl
Postdoctoral supervisor

3. **Paola Pinilla**

Professor, Mullard Space Science Laboratory
University College London
Email: p.pinilla@ucl.ac.uk
Postdoctoral supervisor

4. **Myriam Benisty**

Director, Max Planck Institute for Astronomy
Email: benisty@mpia.de
Active Scientific collaborator